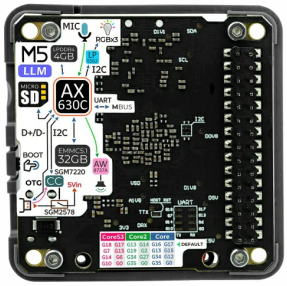
Module LLM

SKU:M140

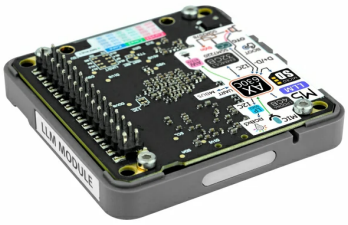


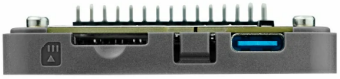




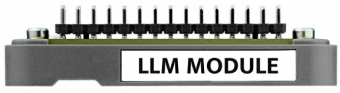














Description

Module LLM is an integrated offline large language model (LLM) inference module, designed for terminal devices that require efficient and intelligent interaction. Whether in smart homes, voice assistants, or industrial control, Module LLM can provide you with a smooth and natural AI experience without relying on the cloud, ensuring privacy, security, and stability.

This module integrates the **StackFlow** framework, and with the **Arduino/UiFlow** library, you can easily achieve edge-side intelligence with just a few lines of code. Equipped with the **AX630C** SoC processor, it integrates a 3.2TOPs high-efficiency NPU, natively supporting Transformer models, and can easily handle complex AI tasks. It comes with **4GB LPDDR4** memory (1GB for user use, 3GB dedicated to hardware acceleration) and **32GB eMMC** storage, supporting parallel loading and serial inference of multiple models, ensuring smooth multitasking. The operating power consumption is only about 1.5W, much lower than similar products, making it energy-efficient and suitable for long-term operation.

Module LLM integrates a microphone, speaker, TF storage card, **USB OTG**, and RGB status light, meeting diverse application needs and easily enabling voice interaction and data transmission. In terms of flexible expansion, the onboard SD card slot supports firmware cold/hot upgrades, and the **UART** communication interface simplifies connection and debugging, ensuring continuous optimization and expansion of module functions. The USB port supports automatic master/slave switching, serving as both a debugging port and allowing connection to more USB devices such as cameras. Paired with the LLM debugging kit, it can expand to a 100M Ethernet port and kernel serial port, functioning as an SBC.

This module is compatible with multiple models and comes pre-installed with the **Qwen2.5-0.5B** large language model, featuring built-in **KWS** (wake word), **ASR** (speech recognition), **LLM** (large language model), and **TTS** (text-to-speech) functions, supporting separate calls or **pipeline** automatic flow, making development convenient. It will later support Qwen2.5-1.5B, Llama3.2-1B, and InternVL2-1B, among other edge-side LLM/VLM models, supporting hot updates to keep up with community trends and adapt to AI tasks of varying complexity. In terms of visual recognition capabilities, it supports CLIP, YoloWorld, and other Open world models, with future updates including DepthAnything, SegmentAnything, and other advanced models, empowering intelligent recognition and analysis.

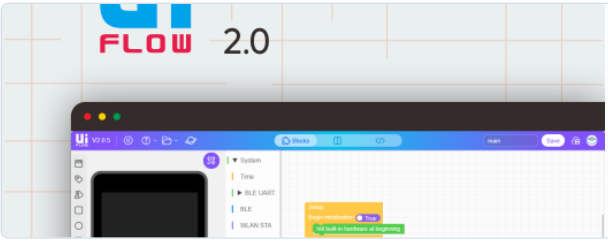
Module LLM is plug-and-play, paired with **M5 hosts**, enabling a plug-and-play AI interaction experience. Users can integrate it into existing smart devices without complicated setups, quickly enabling intelligent functions and enhancing device intelligence. This product is suitable for offline voice assistants, text-to-speech conversion, smart home control, interactive robots, and more.

Tutorial



Arduino IDE

This tutorial will show you how to program and control the Module LLM device using the Arduino IDE



UiFlow2

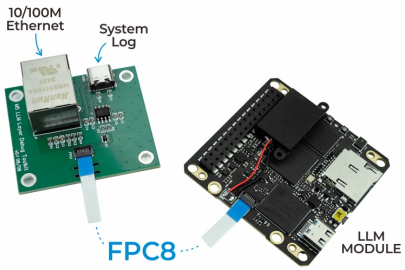
This tutorial will show you how to control the Module LLM device using the UiFlow2 graphical programming platform

Features

- Offline inference, 3.2T@INT8 precision computing power
- Integrated KWS (wake word), ASR (speech recognition), LLM (large language model), TTS (text-to-speech)
- Multiple model parallelism
- Onboard 32GB eMMC storage and 4GB LPDDR4 memory
- Onboard microphone and speaker
- Serial communication
- SD card firmware upgrade
- Supports ADB debugging
- RGB indicator light
- Built-in Ubuntu system
- Supports OTG function
- Supports Arduino/UIFlow

Includes

- 1 x Module LLM



> Debugging board included with the product (limited to initial release)

Applications

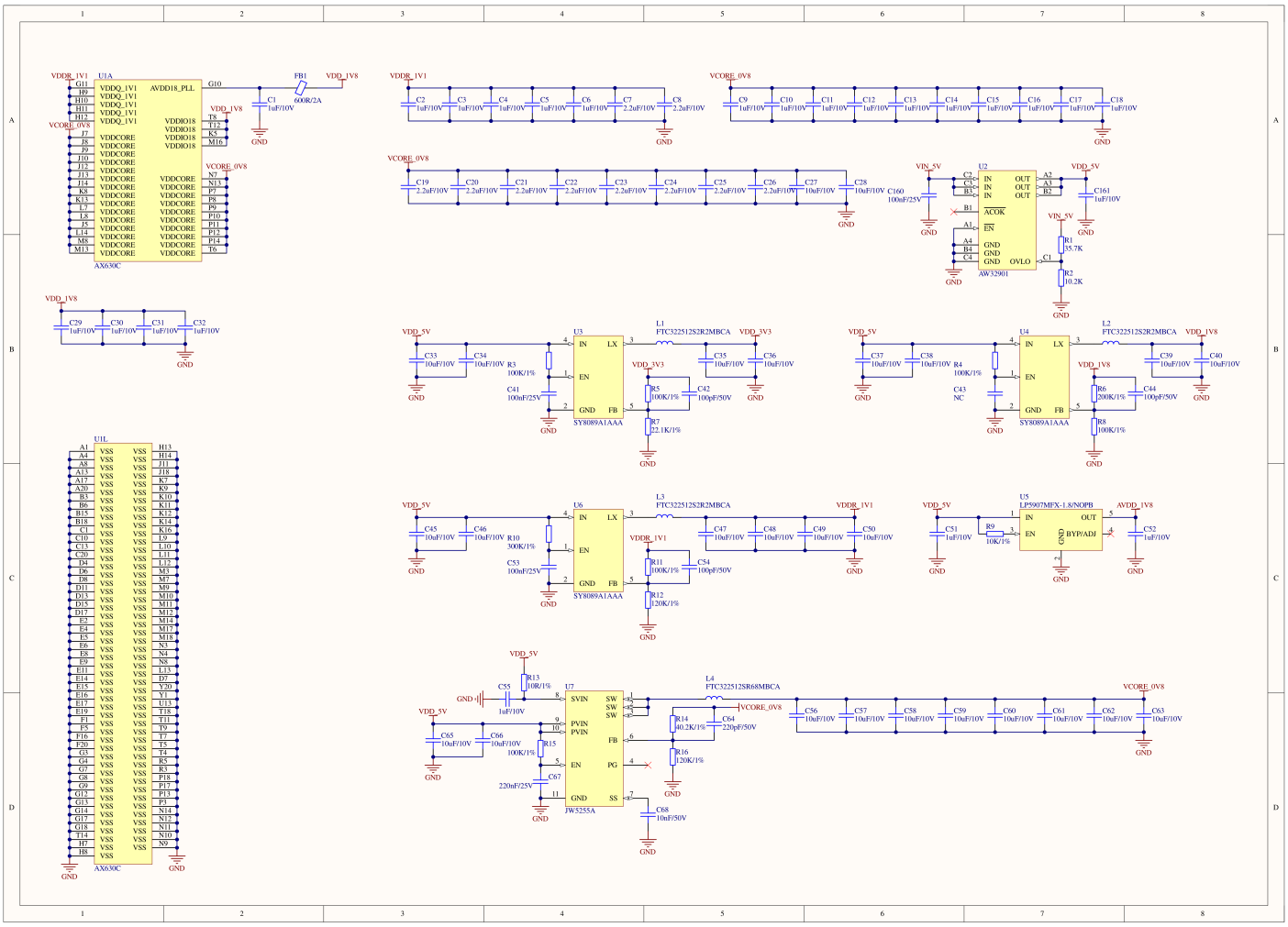
- Offline voice assistant
- Text-to-speech conversion
- Smart home control
- Interactive robots

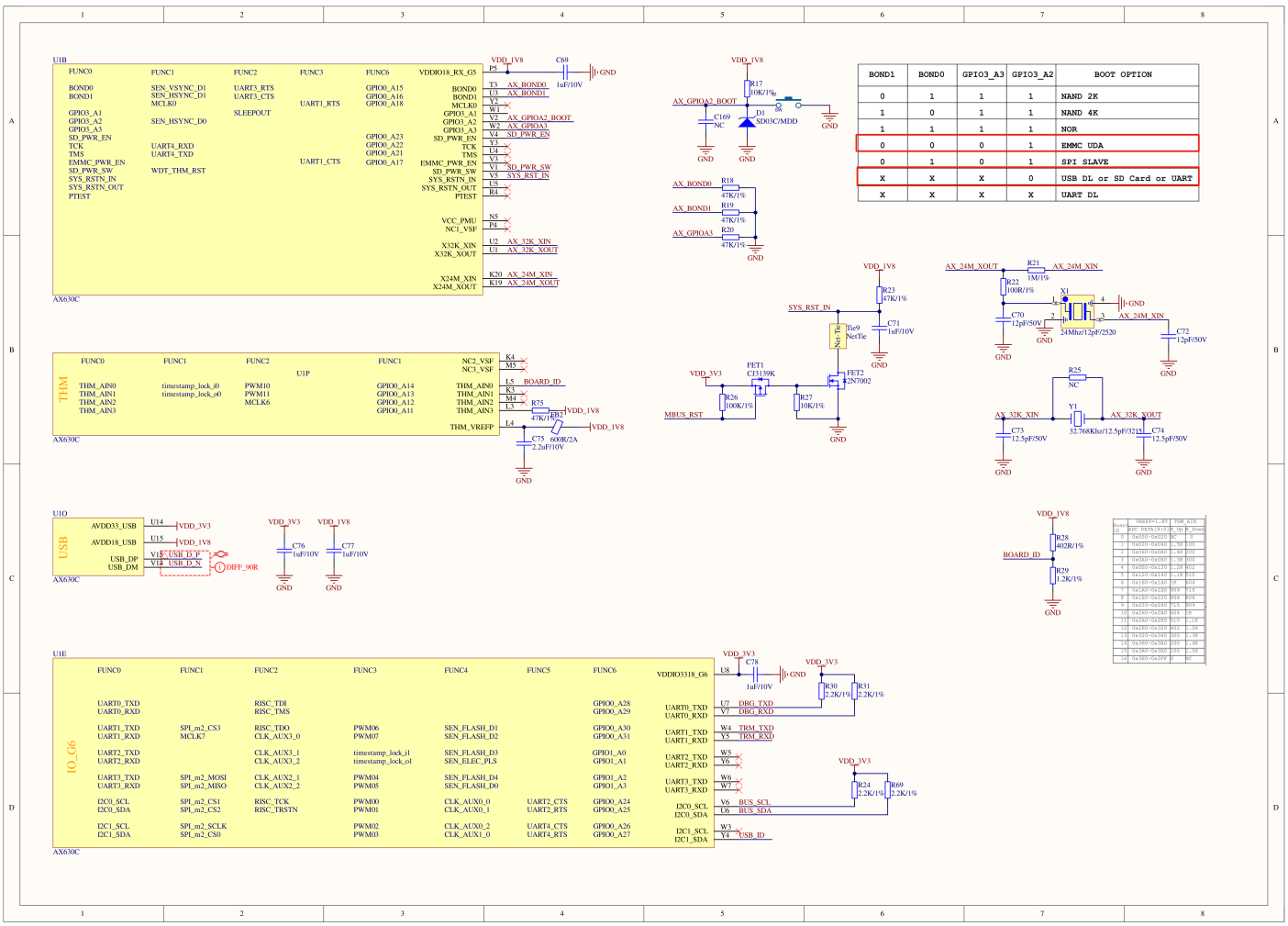
Specifications

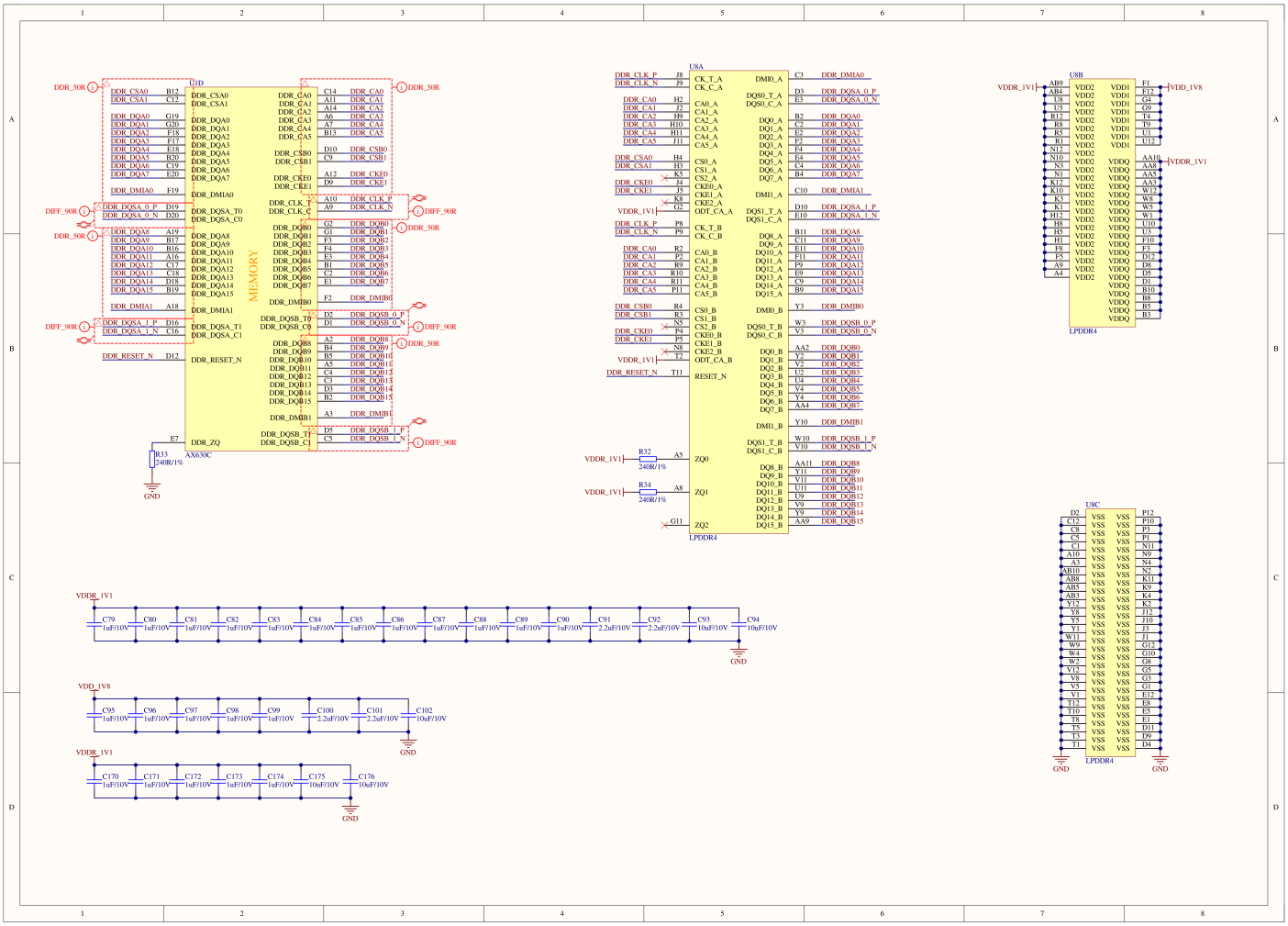
Specification	Parameter
Processor SoC	AX630C@Dual Cortex A53 1.2 GHz MAX.12.8 TOPS @INT4, and 3.2 TOPS @INT8
Memory	4GB LPDDR4 (1GB system memory + 3GB dedicated to hardware acceleration)
Storage	32GB eMMC5.1
Communication Interface	Serial communication, default baud rate 115200@8N1 (adjustable)
Microphone	MSM421A
Audio Driver	AW8737
Speaker	8Ω@1W, size: 2014 cavity speaker
Built-in Functional Units	KWS (wake word), ASR (speech recognition), LLM (large language model), TTS (text-to-speech)
RGB Light	3x RGB LED@2020 driven by LP5562 (status indicator)
Power Consumption	Idle: 5V@0.5W, Full load: 5V@1.5W
Button	Used for entering download mode for firmware upgrade
Upgrade Interface	SD card/Type C port
Operating Temperature	0-40°C
Product Size	54.0 x 54.0 x 13.0mm
Product Weight	17.1g
Package Size	132.0 x 95.0 x 16.0mm
Gross Weight	30.7g

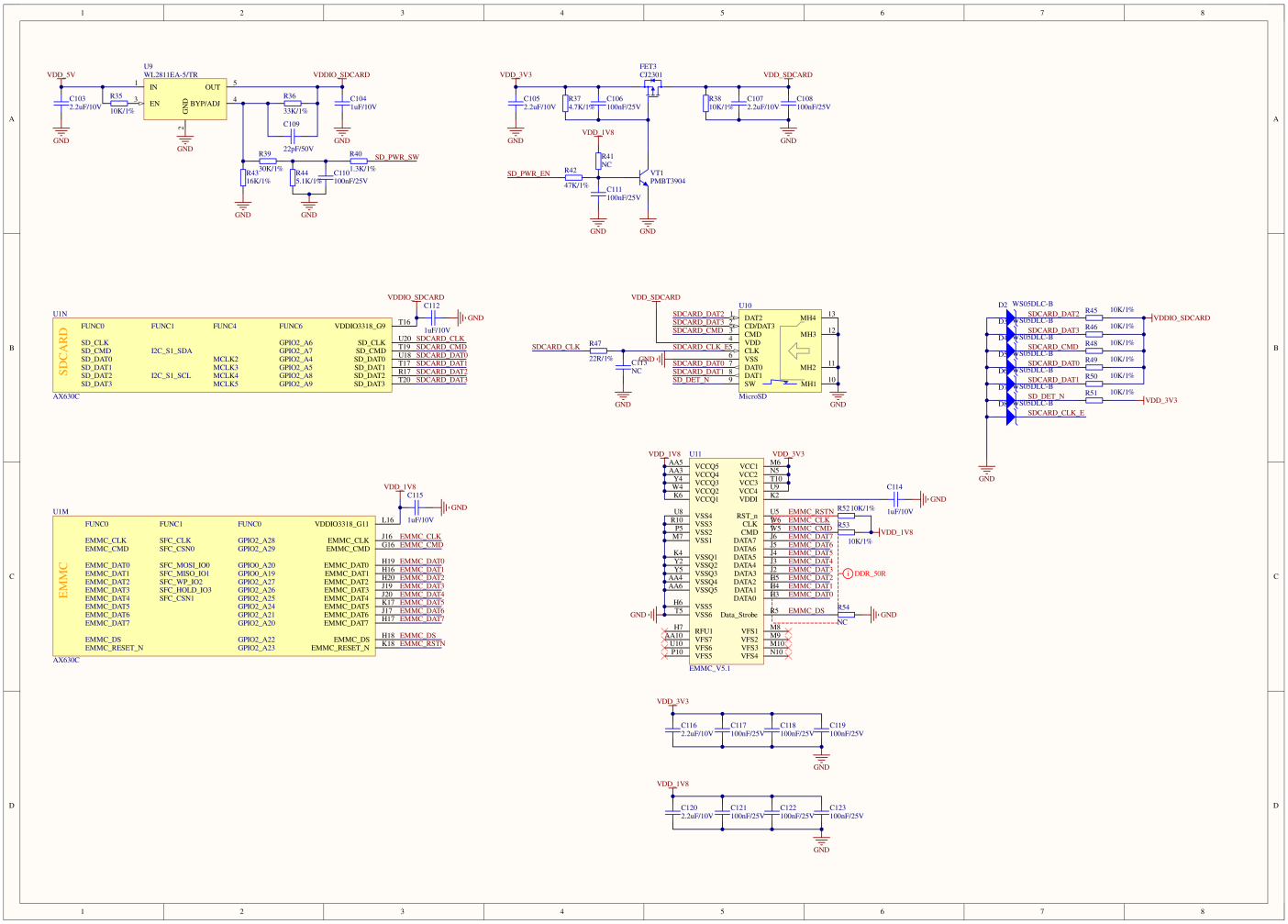
Schematics

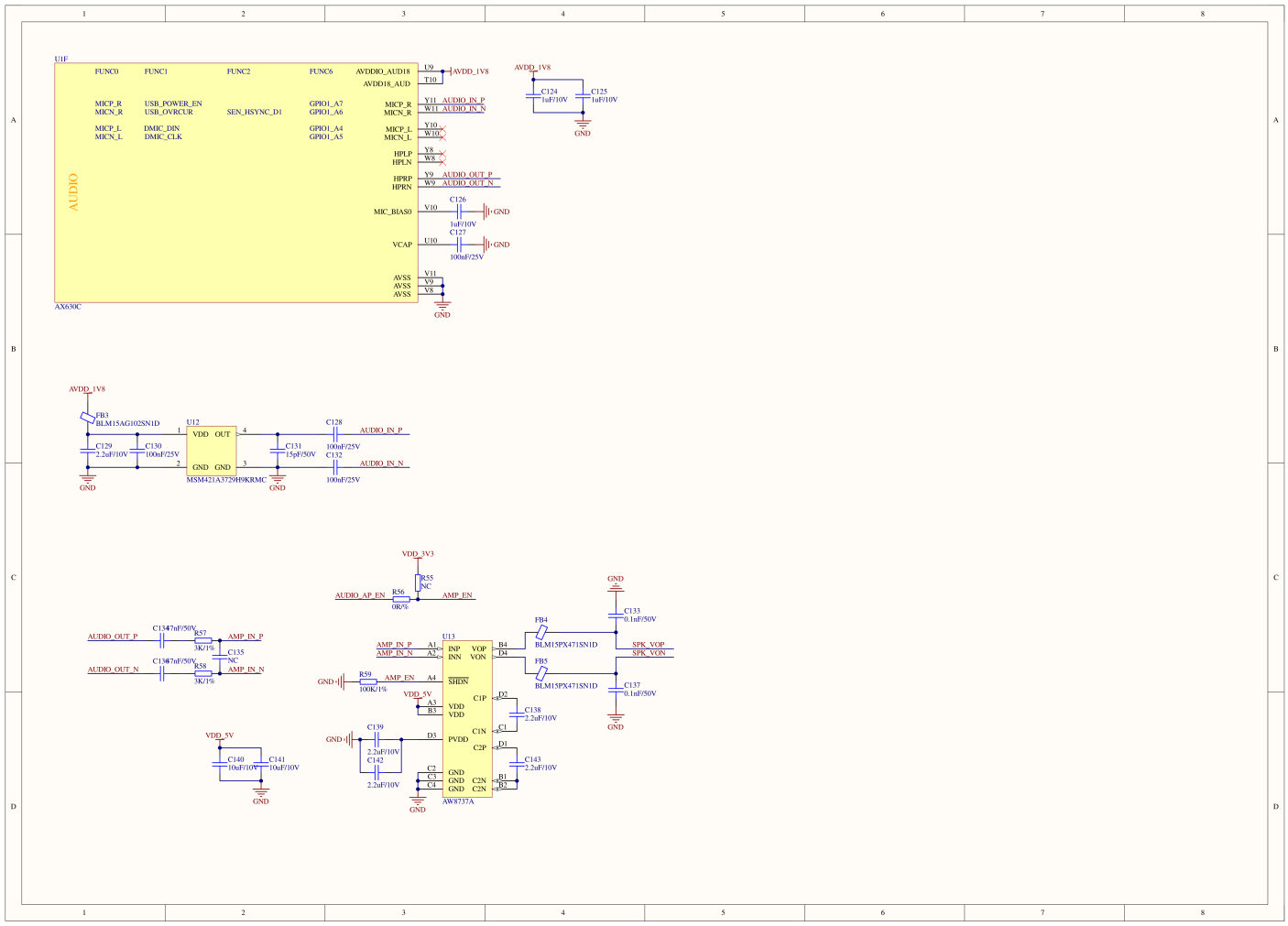
- Module LLM Schematic Download

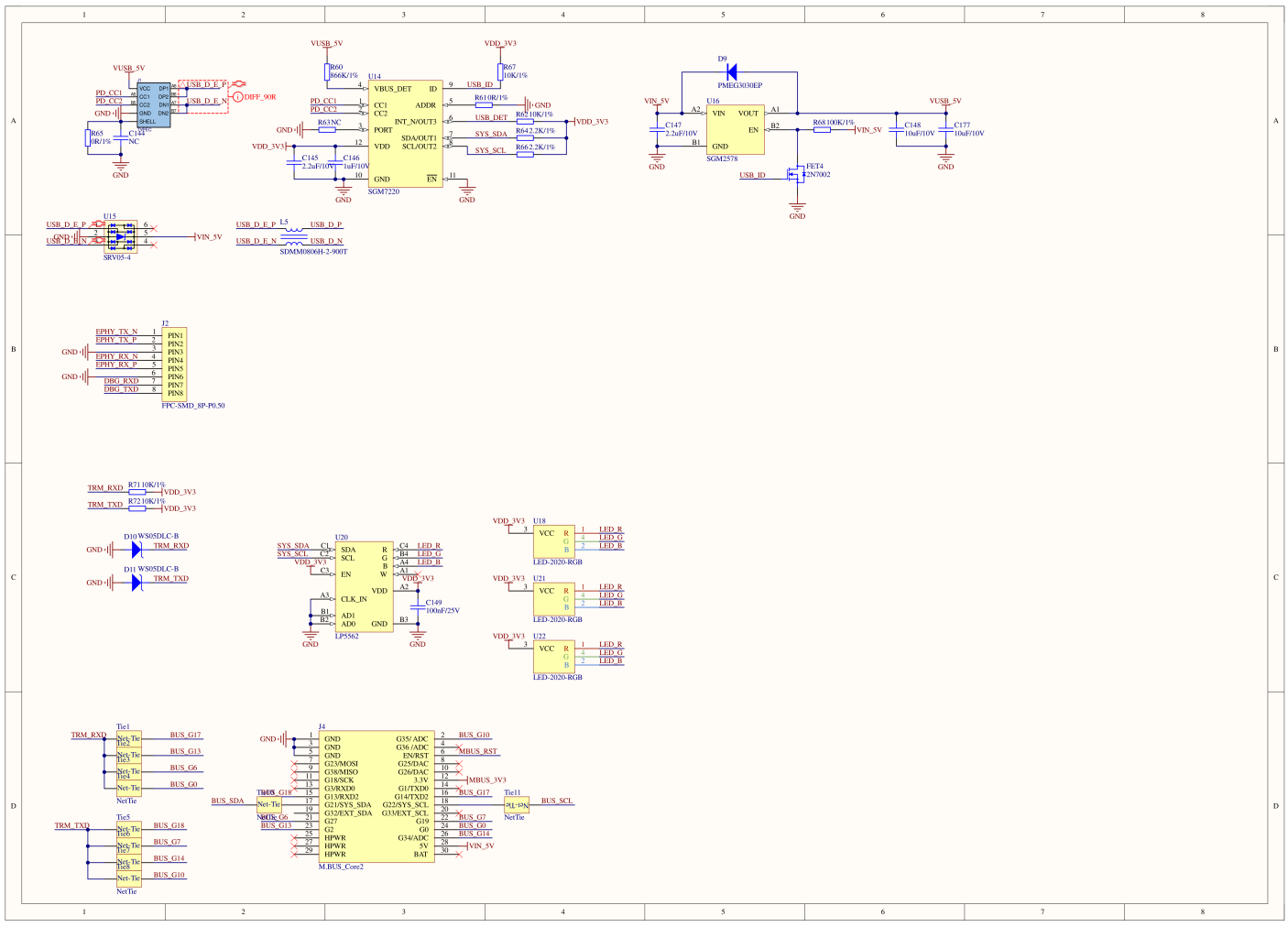

U1A
AX630C
VDDR_1V1
VCORE_0V8
VDD_1V8
VDD_5V
VDD_3V3
VIN_5V
U2
AW32901
U3
SY8089A1AAA
U4
SY8089A1AAA
U6
SY8089A1AAA
U7
JW5255A
U5
LP5907MFX-1.8/NOPB
L1
FTC322512S2R2MBCA
L2
FTC322512S2R2MBCA
L3
FTC322512SR2MBCA
L4
FTC322512SR68MBCA
U1L
GND

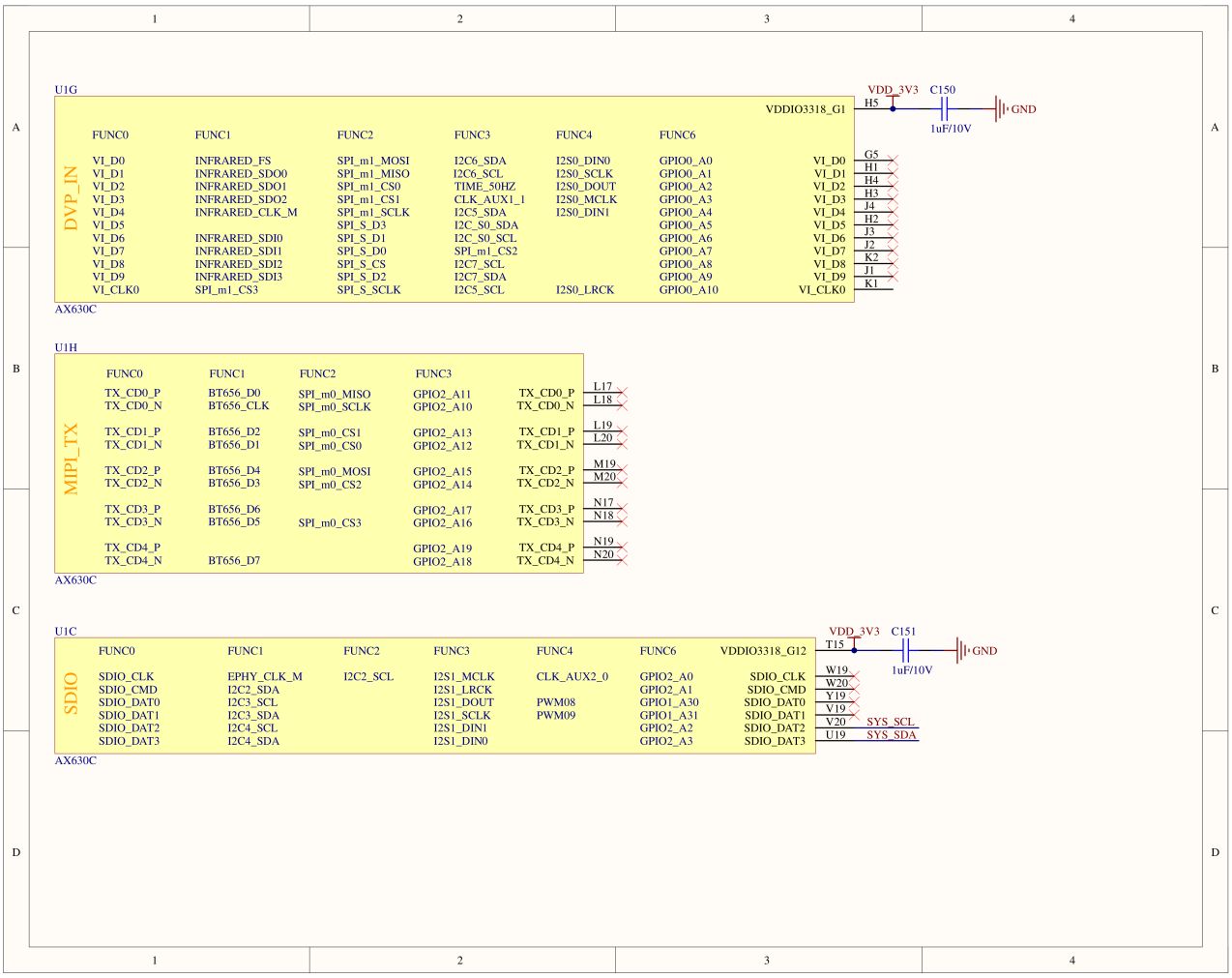

U1B
AX630C
U1P
THM
U1O
USB
U1E
IO_G6
VDD_1V8
VDD_3V3
GND
BOND1
BOND0
GPIO3_A3
GPIO3_A2
BOOT OPTION
0 1 1 1 NAND 2K
1 0 1 1 NAND 4K
1 1 1 1 NOR
0 0 0 1 EMMC UDA
0 1 0 1 SPI SLAVE
X X X 0 USB DL or SD Card or UART
X X X X UART DL

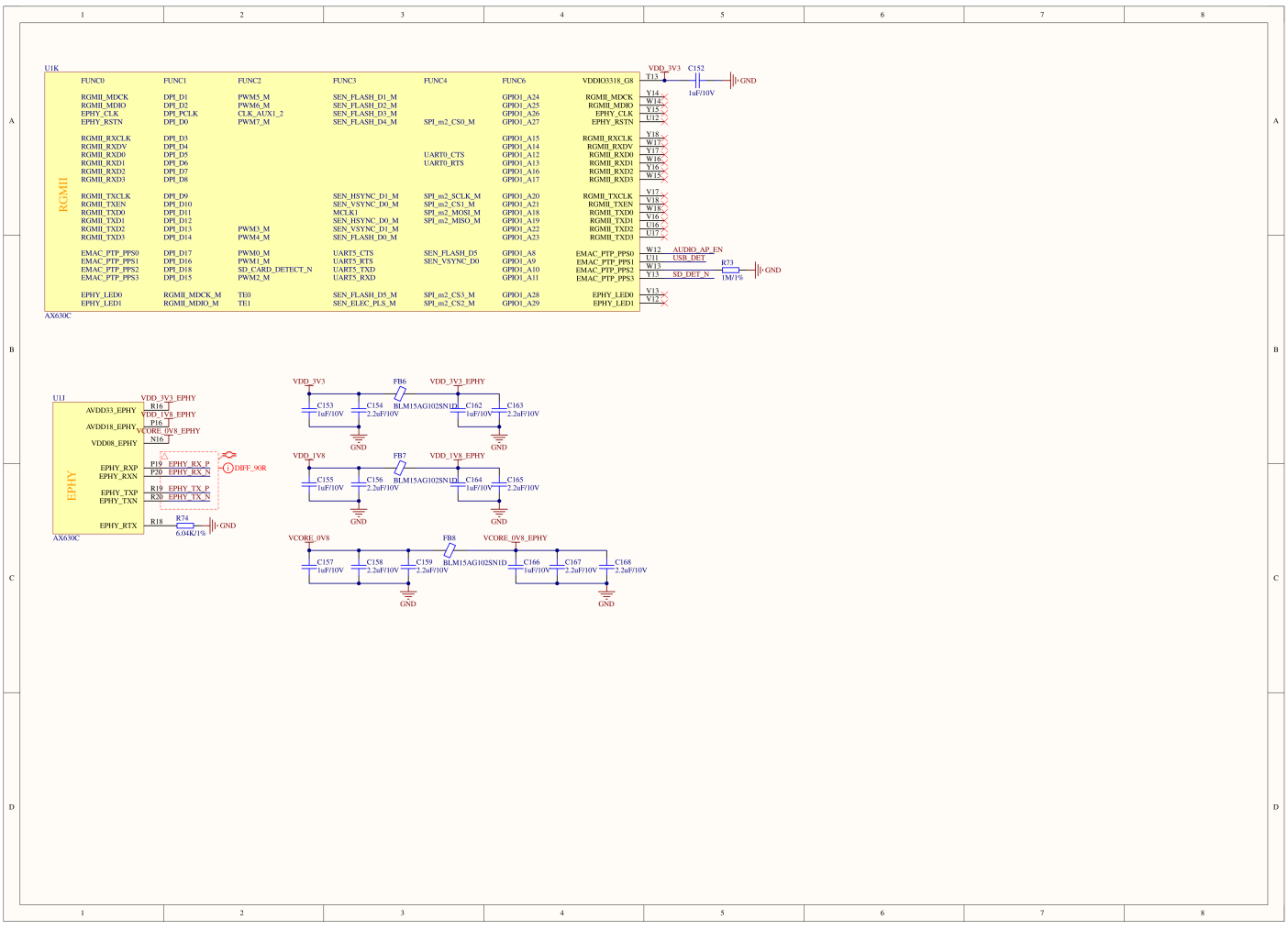

U1D
MEMORY
AX630C
U8A
LPDDR4
U8B
LPDDR4
U8C
LPDDR4
VDD_1V1
VDD_1V8
GND

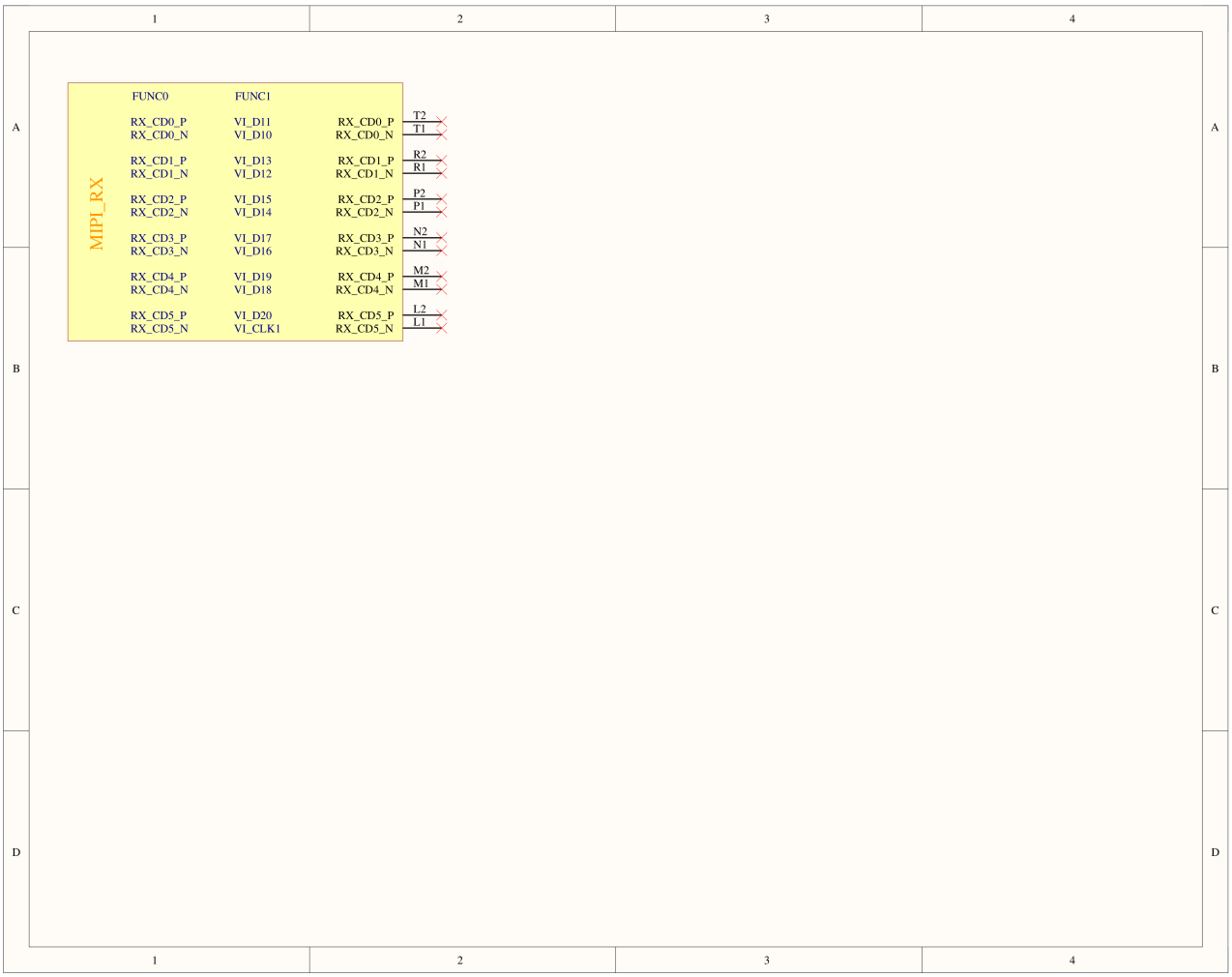

U9
WL2811EA-5/TR
VDD_5V
VDDIO_SDCARD
U1N
SDCARD
AX630C
U10
MicroSD
VDD_SDCARD
VDD_3V3
VDD_1V8
U1M
EMMC
AX630C
U11
EMMC_V5.1
WS05DLC-B
GND


U1F
AUDIO
AX630C
AVDD_1V8
C124 1uF/10V
C125 1uF/10V
GND
C126 1uF/10V
C127 100nF/25V
FB3 BLM15AG102SN1D
U12
MSM421A3729H9KRMC
C128 100nF/25V
C131 15pF/50V
C132 100nF/25V
C129 2.2uF/10V
C130 100nF/25V
AUDIO_IN_P
AUDIO_IN_N
VDD_3V3
R55 NC
R56 0R/1%
AMP_EN
AUDIO_AP_EN
AUDIO_OUT_P
AUDIO_OUT_N
C134 7nF/50V
C135 NC
C136 7nF/50V
R57 3K/1%
R58 3K/1%
AMP_IN_P
AMP_IN_N
U13
AW8737A
R59 100K/1%
C137 0.1nF/50V
C138 2.2uF/10V
C139 2.2uF/10V
C140 10uF/10V
C141 10uF/10V
C142 2.2uF/10V
C143 2.2uF/10V
FB4 BLM15PX471SN1D
FB5 BLM15PX471SN1D
SPK_VOP
SPK_VON


VUSB_5V
VDD_3V3
U14
SGM7220
R60 866K/1%
R67 10K/1%
R610K/1%
R62 10K/1%
R642.2K/1%
R662.2K/1%
USB_ID
SYS_SDA
SYS_SCL
PD_CC1
PD_CC2
R63 NC
C145 2.2uF/10V
C146 1uF/10V
D9
PMEG3030EP
U16
SGM2578
VIN_5V
R68 100K/1%
C147 2.2uF/10V
C148 10uF/10V
C177 10uF/10V
FET4 2N7002
U15
SRV05-4
USB_D_E_P
USB_D_E_N
DIFF_90R
L5
SDMM0806H-2-900IT
USB_D_P
USB_D_N
J2
FPC-SMD_8P-P0.50
EPHY_TX_N
EPHY_TX_P
EPHY_RX_N
EPHY_RX_P
DBG_RXD
DBG_TXD
TRM_RXD
TRM_TXD
R71 10K/1%
R72 10K/1%
D10 WS05DLC-B
D11 WS05DLC-B
U20
LP5562
C149 100nF/25V
U18
U21
U22
LED-2020-RGB
LED_R
LED_G
LED_B
Tie1
Tie5
NetTie
J4
M.BUS_Core2
BUS_G17
BUS_G13
BUS_G6
BUS_G0
BUS_G18
BUS_G7
BUS_G14
BUS_G10
BUS_SDA
BUS_SCL
MBUS_RST
MBUS_3V3
VIN_5V


U1G
VDDIO3318_G1
VDD_3V3
C150
H5
GND
1uF/10V
FUNC0 FUNC1 FUNC2 FUNC3 FUNC4 FUNC6
DVP_IN
VI_D0 INFRARED_FS SPI_m1_MOSI I2C6_SDA I2S0_DIN0 GPIO0_A0 VI_D0 G5
VI_D1 INFRARED_SDO0 SPI_m1_MISO I2C6_SCL I2S0_SCLK GPIO0_A1 VI_D1 H1
VI_D2 INFRARED_SDO1 SPI_m1_CS0 TIME_50HZ I2S0_DOUT GPIO0_A2 VI_D2 H4
VI_D3 INFRARED_SDO2 SPI_m1_CS1 CLK_AUX1_1 I2S0_MCLK GPIO0_A3 VI_D3 H3
VI_D4 INFRARED_CLK_M SPI_m1_SCLK I2C5_SDA I2S0_DIN1 GPIO0_A4 VI_D4 J4
VI_D5 SPI_S_D3 I2C_S0_SDA GPIO0_A5 VI_D5 H2
VI_D6 INFRARED_SDI0 SPI_S_D1 I2C_S0_SCL GPIO0_A6 VI_D6 J3
VI_D7 INFRARED_SDI1 SPI_S_D0 SPI_m1_CS2 GPIO0_A7 VI_D7 J2
VI_D8 INFRARED_SDI2 SPI_S_CS I2C7_SCL GPIO0_A8 VI_D8 K2
VI_D9 INFRARED_SDI3 SPI_S_D2 I2C7_SDA GPIO0_A9 VI_D9 J1
VI_CLK0 SPI_m1_CS3 SPI_S_SCLK I2C5_SCL I2S0_LRCK GPIO0_A10 VI_CLK0 K1
AX630C
U1H
FUNC0 FUNC1 FUNC2 FUNC3
MIPI_TX
TX_CD0_P BT656_D0 SPI_m0_MISO GPIO2_A11 TX_CD0_P L17
TX_CD0_N BT656_CLK SPI_m0_SCLK GPIO2_A10 TX_CD0_N L18
TX_CD1_P BT656_D2 SPI_m0_CS1 GPIO2_A13 TX_CD1_P L19
TX_CD1_N BT656_D1 SPI_m0_CS0 GPIO2_A12 TX_CD1_N L20
TX_CD2_P BT656_D4 SPI_m0_MOSI GPIO2_A15 TX_CD2_P M19
TX_CD2_N BT656_D3 SPI_m0_CS2 GPIO2_A14 TX_CD2_N M20
TX_CD3_P BT656_D6 GPIO2_A17 TX_CD3_P N17
TX_CD3_N BT656_D5 SPI_m0_CS3 GPIO2_A16 TX_CD3_N N18
TX_CD4_P GPIO2_A19 TX_CD4_P N19
TX_CD4_N BT656_D7 GPIO2_A18 TX_CD4_N N20
AX630C
U1C
VDD_3V3
C151
T15
GND
1uF/10V
FUNC0 FUNC1 FUNC2 FUNC3 FUNC4 FUNC6 VDDIO3318_G12
SDIO
SDIO_CLK EPHY_CLK_M I2C2_SCL I2S1_MCLK CLK_AUX2_0 GPIO2_A0 SDIO_CLK W19
SDIO_CMD I2C2_SDA I2S1_LRCK GPIO2_A1 SDIO_CMD W20
SDIO_DAT0 I2C3_SCL I2S1_DOUT PWM08 GPIO1_A30 SDIO_DAT0 Y19
SDIO_DAT1 I2C3_SDA I2S1_SCLK PWM09 GPIO1_A31 SDIO_DAT1 V19
SDIO_DAT2 I2C4_SCL I2S1_DIN1 GPIO2_A2 SDIO_DAT2 V20 SYS_SCL
SDIO_DAT3 I2C4_SDA I2S1_DIN0 GPIO2_A3 SDIO_DAT3 U19 SYS_SDA
AX630C

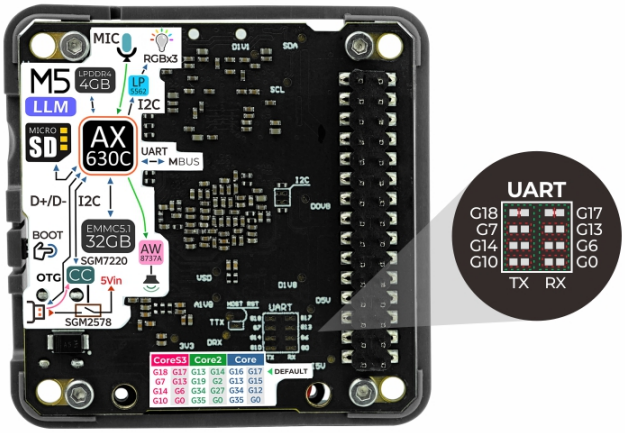

U1K
VDD_3V3 C152
T13
GND
1uF/10V
FUNC0 FUNC1 FUNC2 FUNC3 FUNC4 FUNC6 VDDIO3318_G8
RGMII
RGMII_MDCK DPI_D1 PWM5_M SEN_FLASH_D1_M GPIO1_A24 RGMII_MDCK Y14
RGMII_MDIO DPI_D2 PWM6_M SEN_FLASH_D2_M GPIO1_A25 RGMII_MDIO W14
EPHY_CLK DPI_PCLK CLK_AUX1_2 SEN_FLASH_D3_M GPIO1_A26 EPHY_CLK Y15
EPHY_RSTN DPI_D0 PWM7_M SEN_FLASH_D4_M SPI_m2_CS0_M GPIO1_A27 EPHY_RSTN U12
RGMII_RXCLK DPI_D3 GPIO1_A15 RGMII_RXCLK Y18
RGMII_RXDV DPI_D4 GPIO1_A14 RGMII_RXDV W17
RGMII_RXD0 DPI_D5 UART0_CTS GPIO1_A12 RGMII_RXD0 Y17
RGMII_RXD1 DPI_D6 UART0_RTS GPIO1_A13 RGMII_RXD1 W16
RGMII_RXD2 DPI_D7 GPIO1_A16 RGMII_RXD2 Y16
RGMII_RXD3 DPI_D8 GPIO1_A17 RGMII_RXD3 W15
RGMII_TXCLK DPI_D9 SEN_HSYNC_D1_M SPI_m2_SCLK_M GPIO1_A20 RGMII_TXCLK V17
RGMII_TXEN DPI_D10 SEN_VSYNC_D0_M SPI_m2_CS1_M GPIO1_A21 RGMII_TXEN V18
RGMII_TXD0 DPI_D11 MCLK1 SPI_m2_MOSI_M GPIO1_A18 RGMII_TXD0 W18
RGMII_TXD1 DPI_D12 SEN_HSYNC_D0_M SPI_m2_MISO_M GPIO1_A19 RGMII_TXD1 V16
RGMII_TXD2 DPI_D13 PWM3_M SEN_VSYNC_D1_M GPIO1_A22 RGMII_TXD2 U16
RGMII_TXD3 DPI_D14 PWM4_M SEN_FLASH_D0_M GPIO1_A23 RGMII_TXD3 U17
EMAC_PTP_PPS0 DPI_D17 PWM0_M UART5_CTS SEN_FLASH_D5 GPIO1_A8 EMAC_PTP_PPS0 W12 AUDIO_AP_EN
EMAC_PTP_PPS1 DPI_D16 PWM1_M UART5_RTS SEN_VSYNC_D0 GPIO1_A9 EMAC_PTP_PPS1 U11 USB_DET
EMAC_PTP_PPS2 DPI_D18 SD_CARD_DETECT_N UART5_TXD GPIO1_A10 EMAC_PTP_PPS2 W13
EMAC_PTP_PPS3 DPI_D15 PWM2_M UART5_RXD GPIO1_A11 EMAC_PTP_PPS3 Y13 SD_DET_N
R73
1M/1%
GND
EPHY_LED0 RGMII_MDCK_M TE0 SEN_FLASH_D5_M SPI_m2_CS3_M GPIO1_A28 EPHY_LED0 V13
EPHY_LED1 RGMII_MDIO_M TE1 SEN_ELEC_PLS_M SPI_m2_CS2_M GPIO1_A29 EPHY_LED1 V12
AX630C
U1J
EPHY
VDD_3V3_EPHY
R16
AVDD33_EPHY
VDD_1V8_EPHY
P16
AVDD18_EPHY
VCORE_0V8_EPHY
N16
VDD08_EPHY
EPHY_RXP P19 EPHY_RX_P
EPHY_RXN P20 EPHY_RX_N
DIFF_90R
EPHY_TXP R19 EPHY_TX_P
EPHY_TXN R20 EPHY_TX_N
EPHY_RTX R18 R74 6.04K/1%
GND
AX630C
VDD_3V3 FB6 VDD_3V3_EPHY
C153 1uF/10V C154 2.2uF/10V BLM15AG102SN1D C162 1uF/10V C163 2.2uF/10V
GND GND
VDD_1V8 FB7 VDD_1V8_EPHY
C155 1uF/10V C156 2.2uF/10V BLM15AG102SN1D C164 1uF/10V C165 2.2uF/10V
GND GND
VCORE_0V8 FB8 VCORE_0V8_EPHY
C157 1uF/10V C158 2.2uF/10V C159 2.2uF/10V BLM15AG102SN1D C166 1uF/10V C167 2.2uF/10V C168 2.2uF/10V
GND GND



PinMap

Module LLM Pin Switching

Module LLM reserves pin switching pads. In cases of pin multiplexing conflicts, you can cut the PCB trace and then jumper to another set of pins.



Tx Rx

CoreS3	Core2	Core				
G18	G17	G13	G14	G16	G17	DEFAULT
G7	G13	G19	G2	G13	G15	
G14	G6	G34	G27	G34	G12	
G10	G0	G35	G0	G35	G0	

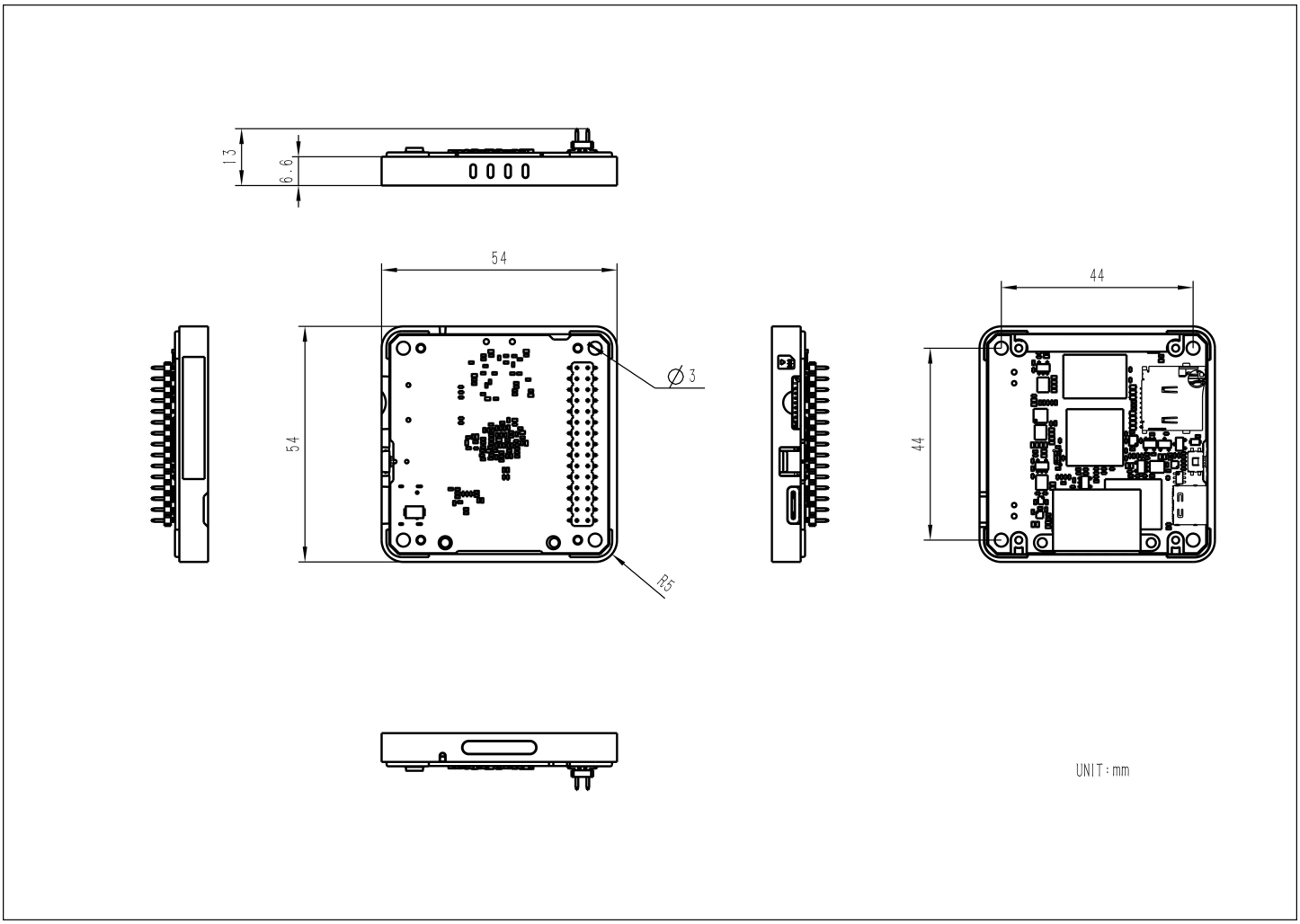
> Taking **CoreS3** as an example, the first column (left green box) is the TX pin for serial communication. Users can choose one of four options (from top to bottom representing pins G18, G7, G14, G10). The default is the IO18 pin. If you need to switch to another pin, cut the pad connection (red line) (recommended to use a blade), and then connect to one of the three pins below. The second column (right green box) is the RX pin switching, as described above, also a four-option operation.

M5-Bus

> **Net-Tie**
>
> The pins marked `NT` in the M5-Bus table below can be switched via the above operation to adapt to different main control devices.

PIN	LEFT	RIGHT	PIN
GND	1	2	TRM_TXD(NT)
GND	3	4	
GND	5	6	RST
	7	8	
	9	10	
	11	12	3V3
	13	14	
TRM_TXD(NT)	15	16	TRM_RXD(NT)
SoC_SDA	17	18	SoC_SCL
	19	20	
TRM_RXD(NT)	21	22	TRM_TXD(NT)
TRM_RXD(NT)	23	24	TRM_RXD(NT)
	25	26	TRM_TXD(NT)
	27	28	5V
	29	30	

Model Size



Datasheets

- AX630C

Softwares

Quick Start

- Module LLM ADB / SSH / UART Connect
- Module LLM Software Update
- Module LLM Image Update
- Module LLM Status Indicator:
  - Red: Device initializing
  - Green: Device initialization complete
- Module LLM Application Upgrade Status Indicator:
  - Blue blinking: Application package updating
  - Red: Application package upgrade failed
  - Green: Application package upgrade successful

Module LLM Model Replacement Notes

The models supported by the Module LLM are in a special format unique to AXERA and require special processing to be used normally. Therefore, existing models on the market cannot be used directly.

Arduino

- Module LLM Arduino Quick Start
- Module LLM Arduino Library
- Module LLM Arduino Library API

UiFlow1

- Module LLM UiFlow1 Docs

UiFlow2

- Module LLM UiFlow2 Quick Start
- Module LLM UiFlow2 Docs
- Module LLM UiFlow2 API

OpenAI API

- Module LLM OpenAI API tutorial

Development Framework

- StackFlow

Development Resources

- AX630C Databrief
- Module LLM JSON API Documentation
- Module LLM AX630C API Docs
- Module LLM AX620E MSP/Sample
- Module LLM Linux Kernel 4.19.125-head
- AXERA LLM Example
- AXERA CV Example
- Module LLM Large Model Compilation Guide

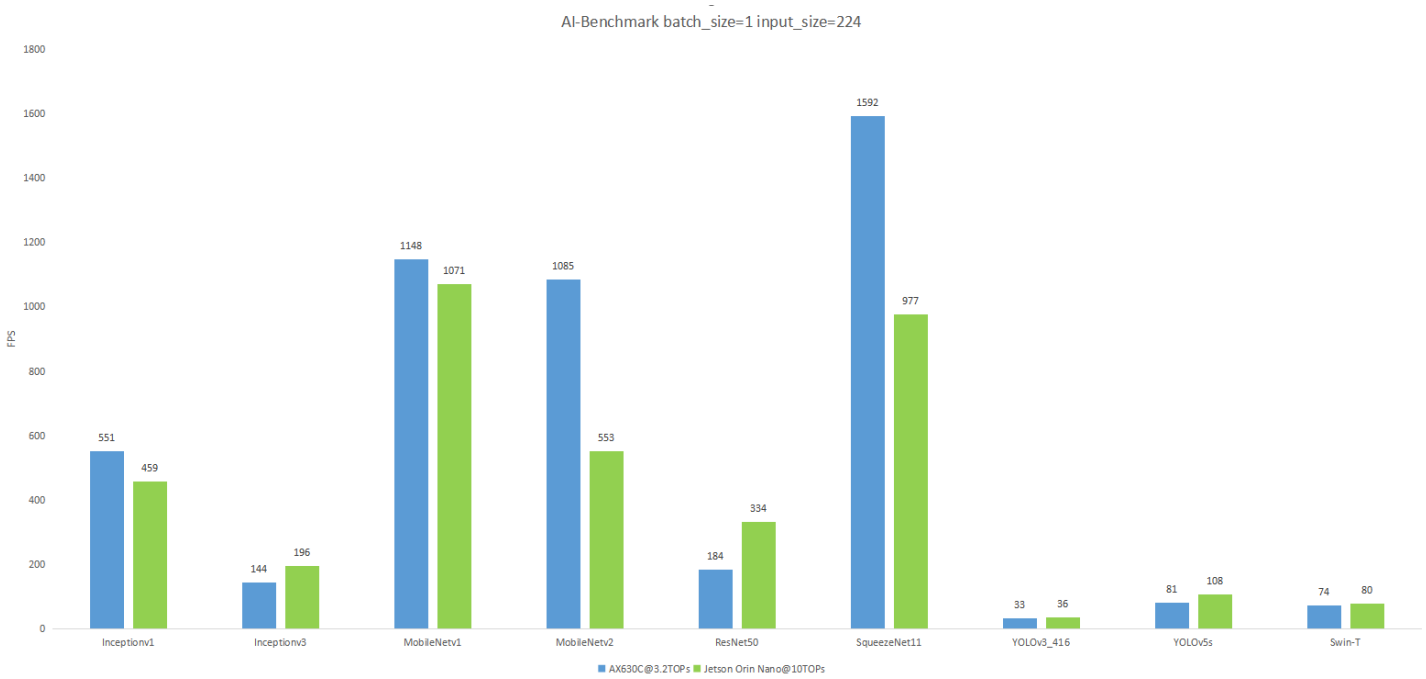
Video

- Module LLM Product Introduction and Example Demonstration

Module_LLM_Video.mp4

Product Comparison

AI Benchmark Comparison


AI-Benchmark batch_size=1 input_size=224
FPS
1800
1600
1400
1200
1000
800
600
400
200
0
551
459
144
196
1148
1071
1085
553
184
334
1592
977
33
36
81
108
74
80
Inceptionv1
Inceptionv3
MobileNetv1
MobileNetv2
ResNet50
SqueezeNet11
YOLOv3_416
YOLOv5s
Swin-T
AX630C@3.2TOPs
Jetson Orin Nano@10TOPs